

SQL CASE statements

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Exercise 03:

Products		
product_id	product_name	price
1	Laptop	1200.00
2	Phone	800.00
3	Keyboard	45.00
4	Monitor	300.00
5	Mouse	25.00

01. Classify each product by price into the three tiers below.

- 'Expensive' if price > 1000
- 'Mid-range' if price between 100 and 1000
- 'Budget' if price < 100

Expected columns: Product_name, price, price_category

```
SELECT product_name,  
       price,  
       CASE  
         WHEN price > 1000 THEN 'Expensive'  
         WHEN price BETWEEN 100 AND 1000 THEN 'Mid-range'  
         ELSE 'Budget'  
       END AS price_category  
FROM products;
```

results		
product_name	price	Price_category
Laptop	1200.00	Expensive
Phone	800.00	Mid-range
Keyboard	45.00	Budget
Monitor	300.00	Mid-range
Mouse	25.00	Budget

orders		
order_id	customer_name	amount
1	Alice	150.00
2	Bob	560.00
3	Charlie	999.99
4	Diana	45.50
5	Ethan	1200.00

Q2. Label each order by its value.

- 'High Value' for order ≥ 1000
- 'Medium Value' for 500 - 999.99
- 'Low Value' for orders < 500

EXPECTED COLUMNS: customer_name, amount, order_value_category

```

SELECT customer_name,
       amount,
CASE
  WHEN amount >= 1000 THEN 'High Value'
  WHEN amount BETWEEN 500 AND 999.99 THEN 'Medium Value'
  ELSE 'Low Value'
END AS order_value_category
FROM orders;

```

results

customer_name	amount	order_value_category
Alice	150.00	Low value
Bob	560.00	Medium value
Charlie	999.99	Medium Value
Diana	45.50	Low value
Ethan	1200.00	High Value

employees			
emp_id	emp_name	department	salary
1	John	IT	85 000
2	Sara	HR	60 000
3	Mark	IT	75 000
4	Lucy	Finance	95 000
5	Tom	HR	55 000

03. Categorize each employee's position level using both department and salary.

- If in 'IT' and salary > 80 000 → 'Senior IT'
- If in 'HR' and salary > 55 000 → 'Experienced HR'
- Otherwise → 'Staff'

EXPECTED COLUMNS: emp_name, department, salary, position_level

```
SELECT emp_name,
       department,
       salary,
```

CASE

WHEN department = 'IT' AND salary > 80000 THEN 'Senior IT'

WHEN department = 'HR' AND salary > 55000 THEN 'Experienced HR'

ELSE 'Staff'

END AS position_level

```
FROM employees;
```

results

emp_name	department	salary	position_level
John	IT	85 000	Senior IT
Sara	HR	60 000	Experienced HR
Mark	IT	75 000	Staff
Lucy	Finance	95 000	Staff
Tom	HR	55 000	Staff

Students		
student_id	student_name	score
1	Anna	92
2	Ben	76
3	Cara	59
4	David	83
5	Ella	68

04. Assign each student a letter grade based on their score.

- $\geq 90 \rightarrow 'A'$
- $80-89 \rightarrow 'B'$
- $70-79 \rightarrow 'C'$
- $60-69 \rightarrow 'D'$
- $< 60 \rightarrow 'F'$

EXPECTED COLUMNS: student_name, score, grade.

SELECT student_name,
score,

CASE

WHEN score ≥ 90 THEN 'A'

WHEN score BETWEEN 80 AND 89 THEN 'B'

WHEN score BETWEEN 70 AND 79 THEN 'C'

WHEN score BETWEEN 60 AND 69 THEN 'D'

ELSE 'F'

END AS grade

FROM students;

results

student_name	score	grade
Anna	92	A
Ben	76	C
Cara	59	F
David	83	B
Ella	68	D

deliveries	
delivery-id	delivery-time_minutes
1	45
2	80
3	30
4	65
5	100

Q5. Label delivery performance based on the time taken.

- ≤ 30 mins $\rightarrow$ 'Fast'
- 31-60 mins $\rightarrow$ 'On Time'
- > 60 mins $\rightarrow$ 'Late'

Expected columns: delivery-id, delivery-time_minutes, performance.

SELECT *

CASE

WHEN delivery-time_minutes ≤ 30 mins THEN 'Fast'

WHEN delivery-time_minutes BETWEEN 31^{mins} AND 60^{mins} THEN 'On Time'

WHEN delivery-time_minutes > 60 mins THEN 'Late'

END AS performance

FROM deliveries;

results		
delivery-id	delivery-time_minutes	performance
1	45	On Time
2	80	Late
3	30	Fast
4	65	Late
5	100	Late

tickets		
ticket_id	issue_type	priority
1	Login issue	1
2	Server down	3
3	Slow system	2
4	Email error	2
5	Password reset	1

Q6. Convert the numeric priority into a readable label.

- 3 → 'High'
- 2 → 'Medium'
- 1 → 'Low'

EXPECTED columns: issue_type, priority, priority_label.

SELECT issue_type,

priority,

CASE

WHEN priority = 3 THEN 'High'

WHEN priority = 2 THEN 'Medium'

ELSE 'Low'

END AS priority_label

FROM tickets;

results		
issue_type	priority	priority_label
Login issue	1	Low
Server down	3	High
Slow system	2	Medium
Email error	2	Medium
Password reset	1	Low

attendance			
student_id	days_present	total_days	attendance_percentage
1	45	50	90%
2	30	50	60%
3	48	50	96%
4	25	50	50%
5	50	50	100%

07. Calculate attendance percentage and classify the result.

- $\geq 90\%$ $\rightarrow$ 'Excellent'
- 75-89% $\rightarrow$ 'Good'
- $< 75\%$ $\rightarrow$ 'Needs Improvement'

EXPECTED columns: student_id, attendance_percentage, attendance_status

```
SELECT student_id,
        attendance_percentage,
        CASE
```

```
    WHEN attendance_percentage  $\geq 90\%$  THEN 'Excellent'
```

```
    WHEN attendance_percentage BETWEEN 75 AND 89% THEN 'Good'
```

```
    ELSE 'Needs Improvement'
```

```
END AS attendance_status
```

```
FROM attendance;
```

student_id	attendance_percentage	attendance_status
1	90%	Excellent
2	60%	Needs Improvement
3	96%	Excellent
4	50%	Need Improvement
5	100%	Excellent

products_inventory	
product_id	stock_qty
1	5
2	0
3	25
4	10
5	3

08. Label the stock status of each product.

• 0 → 'Out of stock'

• 1-5 → 'Low Stock'

• > 5 → 'In stock'

EXPECTED columns: product_id, stock_qty, stock_status

SELECT product_id,
stock_qty,

CASE

WHEN stock_qty = 0 THEN 'Out of stock'

WHEN stock_qty BETWEEN 1 AND 5 THEN 'Low Stock'

ELSE 'In stock'

END AS stock_status

FROM products_inventory;

product_id	stock_qty	stock_status
1	5	Low stock
2	0	Out of stock
3	25	In stock
4	10	In stock
5	3	Low stock

classes		
class_id	subject	enrolled_students
1	Math	30
2	English	25
3	Science	15
4	Art	5
5	History	20

09. Classify each class by the number of enrolled students.

• $\geq 25 \rightarrow$ 'Large'

• $10-24 \rightarrow$ 'Medium'

• $< 10 \rightarrow$ 'Small'

EXPECTED COLUMNS: subject, enrolled_students, class_size_category.

SELECT subject,

enrolled_students,

CASE

WHEN enrolled_students ≥ 25 THEN 'Large'

WHEN enrolled_students BETWEEN 10 AND 24 THEN 'Medium'

ELSE 'Small'

END AS class_size_category

FROM classes;

payment_id	payments	
	amount	payment_method
1	50.00	Card
2	200.00	Cash
3	150.00	Card
4	75.00	Paypal
5	300.00	Cash

10. Apply a discount flag based on the payment method and amount.
- If payment_method = 'cash' and amount $\geq$ 200 $\rightarrow$ 'Eligible for Discount'
 - Otherwise $\rightarrow$ 'Not Eligible'

EXPECTED COLUMNS: payment_id, payment_method, amount, discount_eligibility,

SELECT*

CASE

WHEN payment_method = 'cash' AND amount $\geq$ 200
 THEN 'Eligible for Discount'

ELSE 'Not Eligible'

END AS discount_eligibility =

FROM payments;

results

payment_id	amount	payment_method	discount_eligibility
1	50.00	Card	Not eligible.
2	200.00	Cash	Eligible for Discount
3	150.00	Card	Not Eligible
4	75.00	Paypal	Not eligible
5	300.00	Cash	Eligible for Discount